

## Demonstration of Proposed Features

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|----------------------|---------------------------------------------------------------------|
| <b>Date</b>          | 15 March 2026                                                       |
| <b>Team ID</b>       | XXXXXX                                                              |
| <b>Project Name</b>  | <b>Smart Lender</b> – AI-Powered Loan Eligibility Prediction System |
| <b>Maximum Marks</b> | 1 Mark                                                              |

### Demonstration of Proposed Features:

This document summarizes the key features proposed during the planning phase of the **Smart Lender** project and records their implementation and demonstration status. It serves as evidence that the proposed functionalities were successfully developed, integrated, tested, and demonstrated as part of the final project,

| S.No | Feature Name                            | Description                                                                                                                                                                                                                                                                                     | Status (Implemented / Partial / Pending) | Demonstrated (Yes / No) | Remarks                                                                                                                        |
|------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 1    | Machine Learning Prediction Pipeline    | Developed a complete machine learning pipeline that preprocesses historical loan data, trains four classification algorithms (Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost), evaluates their performance, and automatically selects the best-performing model for deployment. | Implemented                              | Yes                     | Successfully demonstrated model training, evaluation, automatic model selection, and generation of serialized model artifacts. |
| 2    | Interactive Web Application             | Developed a responsive Flask-based web application that allows users to enter applicant information and obtain real-time loan eligibility predictions through an intuitive user interface.                                                                                                      | Implemented                              | Yes                     | Successfully demonstrated real-time prediction using the loan application form and responsive user interface.                  |
| 3    | AI-Based Prediction Result Analysis     | Designed a prediction results page displaying loan approval status, confidence score, AI feature contribution analysis, estimated EMI, Debt-to-Income (DTI) ratio, and complete applicant summary.                                                                                              | Implemented                              | Yes                     | Demonstrated explainable AI through feature importance visualization and prediction confidence indicators.                     |
| 4    | EMI and Debt-to-Income (DTI) Calculator | Implemented integrated financial calculators to compute the applicant's monthly EMI and Debt-to-Income ratio based on loan amount, interest rate, tenure, and monthly income.                                                                                                                   | Implemented                              | Yes                     | Successfully demonstrated automatic EMI and DTI calculations for different loan scenarios.                                     |
| 5    | Batch Prediction System                 | Developed a batch processing module that accepts CSV files containing multiple loan applications, predicts loan eligibility for each applicant, and generates a downloadable prediction report.                                                                                                 | Implemented                              | Yes                     | Demonstrated successful upload, processing, prediction, and download of batch prediction results.                              |
| 6    | Model Performance Dashboard             | Developed a dashboard displaying machine learning evaluation metrics, model comparison charts, confusion matrices, ROC curves, feature                                                                                                                                                          | Implemented                              | Yes                     | Successfully demonstrated visualization of model performance                                                                   |

|  |  |                                                                               |  |  |                                                      |
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|  |  | importance analysis, and other performance reports generated during training. |  |  | reports and comparison among all trained algorithms. |
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**Feature Implementation Summary:**

|                                 |      |
|---------------------------------|------|
| Total Features Proposed         | 6    |
| Total Features Implemented      | 6    |
| Total Features Demonstrated     | 6    |
| Overall Implementation Rate (%) | 100% |